Ratio Test Problems & Solutions

Infinite Series Convergence Handout

Part 1: Problem List

Test for convergence the following infinite series using D'Alembert's Ratio Test:

1. $x + \dfrac{x^2}{2} + \dfrac{x^3}{3} + \ldots \infty$

2. $\sqrt{\dfrac{1}{2}}x + \sqrt{\dfrac{2}{3}}x^2 + \sqrt{\dfrac{3}{4}}x^3 + \ldots \infty$

3. $1 + \dfrac{x}{2} + \dfrac{x^2}{5} + \dfrac{x^3}{10} + \cdots + \dfrac{x^n}{n^2+1} + \ldots \infty$

4. $\sum_{n=3}^{\infty} \dfrac{x^n}{n(n-1)(n-2)}$

5. $1 + \dfrac{2^p}{2!} + \dfrac{3^p}{3!} + \dfrac{4^p}{4!} + \ldots \infty$

6. $\sum_{n=1}^{\infty} \dfrac{n!\,3^n}{n^n}$

7. $\sum_{n=1}^{\infty} \dfrac{n^3 + a}{2^n + a} \quad (a > 0)$

8. $\sum_{n=1}^{\infty} \dfrac{\sqrt{n}}{\sqrt{n^2+1}}\, x^n$

9. $\sum_{n=1}^{\infty} \dfrac{n^3 - n + 1}{n!}$

10. $\left(\dfrac{1}{3}\right)^2 + \left(\dfrac{1\cdot 2}{3\cdot 5}\right)^2 + \left(\dfrac{1\cdot 2\cdot 3}{3\cdot 5\cdot 7}\right)^2 + \ldots \infty$

11. $1 + \dfrac{1^2\cdot 2^2}{1\cdot 3\cdot 5} + \dfrac{1^2\cdot 2^2\cdot 3^2}{1\cdot 3\cdot 5\cdot 7\cdot 9} + \ldots \infty$

12. $\dfrac{4}{18} + \dfrac{4\cdot 12}{18\cdot 27} + \dfrac{4\cdot 12\cdot 20}{18\cdot 27\cdot 36} + \ldots \infty$

13. $\dfrac{1}{1^p} + \dfrac{x}{3^p} + \dfrac{x^2}{5^p} + \cdots + \dfrac{x^{n-1}}{(2n-1)^p} + \ldots \infty$

14. $\sum_{n=1}^{\infty} \dfrac{3\cdot 6\cdot 9 \ldots 3n}{4\cdot 7\cdot 10 \ldots (3n+1)} \cdot \dfrac{5^n}{3n+2}$

15. $1+\dfrac{1+\alpha}{1+\beta}+\dfrac{(1+\alpha)(1+2\alpha)}{(1+\beta)(1+2\beta)}+\dfrac{(1+\alpha)(1+2\alpha)(1+3\alpha)}{(1+\beta)(1+2\beta)(1+3\beta)}+\ldots\infty \quad (\alpha,\beta>0)$

Part 2: Step-by-Step Solutions

Solution 1.: **General term:** $u_n = \dfrac{x^n}{n}$.

Ratio: $\lim_{n\to\infty} \dfrac{u_{n+1}}{u_n} = \lim_{n\to\infty} \dfrac{x^{n+1}}{n+1} \cdot \dfrac{n}{x^n} = x \lim_{n\to\infty} \dfrac{n}{n+1} = x.$

Conclusion: Converges for $x < 1$, diverges for $x > 1$. At $x = 1$, $u_n = \frac{1}{n}$ (divergent harmonic series). Thus, **converges for** $x < 1$ **and diverges for** $x \geq 1$.

Solution 2.: **General term:** $u_n = \sqrt{\dfrac{n}{n+1}}x^n$.

Ratio: $\lim_{n\to\infty} \dfrac{u_{n+1}}{u_n} = x \lim_{n\to\infty} \sqrt{\dfrac{n+1}{n+2} \cdot \dfrac{n+1}{n}} = x.$

Conclusion: At $x = 1$, $\lim_{n\to\infty} u_n = 1 \neq 0$ (fails n-th term test). Thus, **converges for** $x < 1$ **and diverges for** $x \geq 1$.

Solution 3.: **General term:** $u_n = \dfrac{x^n}{n^2+1}$.

Ratio: $\lim_{n\to\infty} \dfrac{u_{n+1}}{u_n} = x \lim_{n\to\infty} \dfrac{n^2+1}{(n+1)^2+1} = x.$

Conclusion: At $x = 1$, $u_n = \frac{1}{n^2+1} < \frac{1}{n^2}$ (convergent p-series). Thus, **converges for** $x \leq 1$ **and diverges for** $x > 1$.

Solution 4.: **General term:** $u_n = \dfrac{x^n}{n(n-1)(n-2)}$.

Ratio: $\lim_{n\to\infty} \dfrac{u_{n+1}}{u_n} = x \lim_{n\to\infty} \dfrac{n-2}{n+1} = x.$

Conclusion: At $x = 1$, $u_n \sim \frac{1}{n^3}$ ($p = 3 > 1$). Thus, **converges for** $x \leq 1$ **and diverges for** $x > 1$.

Solution 5.: **General term:** $u_n = \dfrac{n^p}{n!}$.

Ratio: $\lim_{n\to\infty} \dfrac{u_{n+1}}{u_n} = \lim_{n\to\infty} \left(\dfrac{n+1}{n}\right)^p \cdot \dfrac{1}{n+1} = 1^p \cdot 0 = 0.$

Conclusion: Since $L = 0 < 1$, the series **converges for all real** p.

Solution 6.: **General term:** $u_n = \dfrac{n!\,3^n}{n^n}$.

Ratio: $\lim_{n\to\infty} \dfrac{u_{n+1}}{u_n} = \lim_{n\to\infty} \dfrac{3}{\left(1+\frac{1}{n}\right)^n} = \dfrac{3}{e} \approx 1.1036.$

Conclusion: Since $L = \frac{3}{e} > 1$, the series **diverges**.

Solution 7.: **General term:** $u_n = \dfrac{n^3+a}{2^n+a} \quad (a > 0).$

Ratio: $\lim_{n\to\infty} \dfrac{u_{n+1}}{u_n} = \lim_{n\to\infty} \dfrac{(n+1)^3+a}{n^3+a} \cdot \dfrac{2^n+a}{2^{n+1}+a} = 1 \cdot \dfrac{1}{2} = \dfrac{1}{2}.$

Conclusion: Since $L = \frac{1}{2} < 1$, the series **converges**.

Solution 8.: **General term:** $u_n = \dfrac{\sqrt{n}}{\sqrt{n^2+1}}\, x^n$.

Ratio: $\lim_{n\to\infty} \dfrac{u_{n+1}}{u_n} = x.$
Conclusion: Converges for $x < 1$ **and diverges for** $x \geq 1$.

Solution 9.: **General term:** $u_n = \dfrac{n^3 - n + 1}{n!}$.
Ratio: $\lim_{n\to\infty} \dfrac{u_{n+1}}{u_n} = \lim_{n\to\infty} \dfrac{(n+1)^3 - (n+1) + 1}{n^3 - n + 1} \cdot \dfrac{1}{n+1} = 1 \cdot 0 = 0.$
Conclusion: Since $L = 0 < 1$, the series **converges**.

Solution 10.: **General term:** $u_n = \left[\dfrac{n!}{3 \cdot 5 \cdot 7 \ldots (2n+1)}\right]^2$.
Ratio: $\lim_{n\to\infty} \dfrac{u_{n+1}}{u_n} = \lim_{n\to\infty} \left(\dfrac{n+1}{2n+3}\right)^2 = \left(\dfrac{1}{2}\right)^2 = \dfrac{1}{4}.$
Conclusion: Since $L = \frac{1}{4} < 1$, the series **converges**.

Solution 11.: **General term:** $u_n = \dfrac{1^2 \cdot 2^2 \ldots (n+1)^2}{1 \cdot 3 \cdot 5 \ldots (4n+1)}$.
Ratio: $\lim_{n\to\infty} \dfrac{u_{n+1}}{u_n} = \lim_{n\to\infty} \dfrac{(n+2)^2}{(4n+3)(4n+5)} = \dfrac{1}{16}.$
Conclusion: Since $L = \frac{1}{16} < 1$, the series **converges**.

Solution 12.: **General term:** $u_n = \dfrac{4 \cdot 12 \cdot 20 \ldots (8n-4)}{18 \cdot 27 \cdot 36 \ldots (9n+9)}$.
Ratio: $\lim_{n\to\infty} \dfrac{u_{n+1}}{u_n} = \lim_{n\to\infty} \dfrac{8n+4}{9n+18} = \dfrac{8}{9}.$
Conclusion: Since $L = \frac{8}{9} < 1$, the series **converges**.

Solution 13.: **General term:** $u_n = \dfrac{x^{n-1}}{(2n-1)^p}$.
Ratio: $\lim_{n\to\infty} \dfrac{u_{n+1}}{u_n} = x.$
Conclusion: Converges for $x < 1$ (for all p), **diverges for** $x > 1$, and at $x = 1$ converges if $p > 1$.

Solution 14.: **General term:** $u_n = \dfrac{3 \cdot 6 \cdot 9 \ldots 3n}{4 \cdot 7 \cdot 10 \ldots (3n+1)} \cdot \dfrac{5^n}{3n+2}$.
Ratio: $\lim_{n\to\infty} \dfrac{u_{n+1}}{u_n} = 5 \lim_{n\to\infty} \dfrac{(3n+3)(3n+2)}{(3n+4)(3n+5)} = 5.$
Conclusion: Since $L = 5 > 1$, the series **diverges**.

Solution 15.: **General term:** $u_n = \dfrac{(1+\alpha)(1+2\alpha)\ldots(1+n\alpha)}{(1+\beta)(1+2\beta)\ldots(1+n\beta)} \quad (\alpha, \beta > 0).$
Ratio: $\lim_{n\to\infty} \dfrac{u_{n+1}}{u_n} = \lim_{n\to\infty} \dfrac{1+(n+1)\alpha}{1+(n+1)\beta} = \dfrac{\alpha}{\beta}.$
Conclusion: Converges if $\beta > \alpha$ $(L < 1)$; **diverges if** $\beta \leq \alpha$ $(L \geq 1)$.